

ROUGH WORK ON MULTIPLE-CHOICE QUESTIONS

QUESTION 1

1.1.

INSTRUCTION	K	A	K >= 0	OUTPUT
DECLARE K, A AS INTEGER	0	0		
FOR K = 3 TO 0 STEP -1	3		T	
A = K + 2		5		
NEXT K	2			
FOR K = 3 TO 0 STEP -1			T	
A = K + 2		4		
NEXT K	1			
FOR K = 3 TO 0 STEP -1			T	
A = K + 2		3		
NEXT K	0			
FOR K = 3 TO 0 STEP -1			T	
A = K + 2		2		
NEXT K	-1			
FOR K = 3 TO 0 STEP -1			F	
DISPLAY A, K				2 -1

Answer:

e. 2 -1

1.2.

INSTRUCTION	K	A	K >= 1	OUTPUT
DECLARE K, A AS INTEGER	0	0		
FOR K = 8 TO 1 STEP -2	8		T	
A = K + 4		12		
NEXT K	6			
FOR K = 8 TO 1 STEP -2			T	
A = K + 4		10		
NEXT K	4			
FOR K = 8 TO 1 STEP -2			T	
A = K + 4		8		
NEXT K	2			
FOR K = 8 TO 1 STEP -2			T	
A = K + 4		6		
NEXT K	0			
FOR K = 8 TO 1 STEP -2			F	
DISPLAY K, A				0 6

Answer:

a. 0 6

1. 3.

INSTRUCTION	K	A	K >= 1	OUTPUT
DECLARE K, A AS INTEGER	0	0		
FOR K = 5 TO 1 STEP -3	5		T	
A = K + A		5		
NEXT K	2			
FOR K = 5 TO 1 STEP -3			T	
A = K + A		7		

NEXT K	-1			
FOR K = 5 TO 1 STEP -3			F	
DISPLAY A, K				7 -1

Answer:

d. none of the above

1.4.

INSTRUCTION	K	A	K >= 2	OUTPUT
DECLARE K, A AS INTEGER	0	0		
FOR K = 4 TO 2 STEP -3	4		T	
A = K + A		4		
NEXT K	1			
FOR K = 4 TO 2 STEP -3			F	
DISPLAY K, A				1 4

Answer:

b. 1 4

QUESTION 2

2.1.

INSTRUCTION	J	TOTAL1	TOTAL2	J <= 3	OUTPUT
DECLARE TOTAL1, TOTAL2, J AS INTEGER	0	0	0		
FOR J = TOTAL1 TO 3 STEP 2	0			T	
TOTAL2 = TOTAL2 + 1			1		
NEXT J	2				
FOR J = TOTAL1 TO 3 STEP 2				T	
TOTAL2 = TOTAL2 + 1			2		

NEXT J	4				
FOR J = TOTAL1 TO 3 STEP 2				F	
DISPLAY TOTAL2					2

Answer:

b. FALSE

2.2.

INSTRUCTION	J	TOTAL1	TOTAL2	J <= 4	OUTPUT
DECLARE TOTAL1, TOTAL2, J AS INTEGER	0	0	0		
FOR J = TOTAL1 TO 4 STEP 2	0			T	
TOTAL2 = TOTAL2 + J			0		
NEXT J	2				
FOR J = TOTAL1 TO 4 STEP 2				T	
TOTAL2 = TOTAL2 + J			2		
NEXT J	4				
FOR J = TOTAL1 TO 4 STEP 2				T	
TOTAL2 = TOTAL2 + J			6		
NEXT J	6				
FOR J = TOTAL1 TO 4 STEP 2				F	
DISPLAY TOTAL2					6

Answer:

a. TRUE

2.3.

INSTRUCTION	J	TOTAL1	TOTAL2	J <= 4	OUTPUT
DECLARE TOTAL1, TOTAL2, J AS INTEGER	0	0	0		
FOR J = TOTAL1 TO 4 STEP 2	0			T	
TOTAL2 = TOTAL2 - J			0		
NEXT J	2				
FOR J = TOTAL1 TO 4 STEP 2				T	
TOTAL2 = TOTAL2 - J			-2		
NEXT J	4				
FOR J = TOTAL1 TO 4 STEP 2				T	
TOTAL2 = TOTAL2 - J			-6		
NEXT J	6				
FOR J = TOTAL1 TO 4 STEP 2				F	
DISPLAY TOTAL2					-6

Answer:

b. FALSE

2.4.

INSTRUCTION	J	TOTAL1	TOTAL2	J <= 4	OUTPUT
DECLARE TOTAL1, TOTAL2, J AS INTEGER	0	0	0		
FOR J = TOTAL1 TO 4 STEP 2	0			T	
TOTAL2 = J - 1			-1		
NEXT J	2				
FOR J = TOTAL1 TO 4 STEP 2				T	
TOTAL2 = J - 1			1		
NEXT J	4				

FOR J = TOTAL1 TO 4 STEP 2				T	
TOTAL2 = J - 1			3		
NEXT J	6				
FOR J = TOTAL1 TO 4 STEP 2				F	
DISPLAY TOTAL2					3

Answer:

a. TRUE

QUESTION 3

3.1.

INSTRUCTION	NUM	SUM	SUM - NUM > 4 AND NUM < SUM	OUTPUT
DECLARE NUM, SUM AS REAL	0	0		
NUM = 3	3			
SUM = NUM ^ 2		9		
IF SUM - NUM > 4 AND NUM < SUM THEN			T	
DISPLAY SUM				9
NUM = SUM * 2	18			

Answer:

b. 9

3.2.

INSTRUCTION	NUM	SUM	SUM - NUM > 4 AND NUM < SUM	OUTPUT
DECLARE NUM, SUM AS REAL	0	0		
NUM = 3	3			
SUM = NUM ^ 2		9		
IF SUM - NUM > 4 AND NUM < SUM THEN			T	
DISPLAY NUM				3
SUM = SUM * 2		18		

Answer:

a. 3

3.3.

INSTRUCTION	NUM	SUM	SUM - NUM < 4 OR NUM < SUM	OUTPUT
DECLARE NUM, SUM AS REAL	0	0		
NUM = 3	3			
SUM = NUM * 2		6		
IF SUM - NUM < 4 OR NUM < SUM THEN			T	
DISPLAY SUM				6
NUM = SUM * 3	18			

Answer:

b. 6

3.4.

INSTRUCTION	NUM	SUM	SUM - NUM < 4 OR NUM < SUM	OUTPUT
DECLARE NUM, SUM AS REAL	0	0		
NUM = 3	3			
SUM = NUM + 2		5		
IF SUM - NUM < 4 OR NUM < SUM THEN			T	
DISPLAY NUM				3
SUM = SUM * 3		15		

Answer:

c. 3

QUESTION 4

4.1 Answer: b. False

Reason: The expression is logically and syntactically wrong; to test “between 21 and 25” you must compare the variable on both sides (e.g., AGE > 21 AND AGE < 25).

4.2 Answer: b. False

Reason: Using OR makes the condition almost always true (AGE >= 20 OR AGE <= 25), so it does not restrict AGE to the range 20–25; the correct connector is AND.

4.3 Answer: b. False

Reason: AGE > 25 AND AGE < 25 is impossible to satisfy (a value cannot be both greater than and less than 25), and it also uses the wrong range for 20–25.

4.4 Answer: b. False

Reason: AGE >= 20 AND AGE <= 25 checks 20–25 inclusive, not 21–25; the lower bound should be >= 21 for 21–25.

QUESTION 5

5.1 Answer: b. False

Reason: The code excludes BEDS = 200 for no stated reason and leaves TOTALBEDS unused; also CNT1 must be properly initialized and named consistently to hold the accumulated total.

5.2 Answer: b. False

Reason: $CNT1 = CNT1 + 1$ counts entries not equal to 200 rather than accumulating the number of beds, so it does not compute the total beds.

5.3 Answer: a. True

Reason: IF BEDS > 0 THEN $CNT1 = CNT1 + BEDS$ correctly adds each positive BEDS value into CNT1, producing the total.

5.4 Answer: b. False

Reason: $CNT1 = CNT1 + 1$ inside the IF counts how many positive entries there are, not the sum of beds, so it does not accumulate total beds.

QUESTION 6

6.1.

INSTRUCTION	K	PENS	CNT1	K <= 3	PENS > 8 OR PENS < 10	OUTPUT
DECLARE K, PENS, TOTAL, CNT1 AS INTEGER	0	0	0			
FOR K = 1 TO 3	1			T		
ACCEPT PENS		8				
IF PENS > 8 OR PENS < 10 THEN					T	
CNT1 = CNT1 + 1			1			
NEXT K	2					
FOR K = 1 TO 3				T		
ACCEPT PENS		10				
IF PENS > 8 OR PENS < 10 THEN					T	
CNT1 = CNT1 + 1			2			
NEXT K	3					
FOR K = 1 TO 3				T		
ACCEPT PENS		9				
IF PENS > 8 OR PENS < 10 THEN					T	
CNT1 = CNT1 + 1			3			
NEXT K	4					
FOR K = 1 TO 3				F		
DISPLAY CNT1						3

Answer:

c. 3

6.2.

INSTRUCTION	K	PENS	CNT1	K <= 3	PENS > 8 AND PENS < 10	OUTPUT
DECLARE K, PENS, TOTAL, CNT1 AS INTEGER	0	0	0			
FOR K = 1 TO 3	1			T		
ACCEPT PENS		8				
IF PENS > 8 AND PENS < 10 THEN					F	
NEXT K	2					
FOR K = 1 TO 3				T		
ACCEPT PENS		10				
IF PENS > 8 AND PENS < 10 THEN					F	
NEXT K	3					
FOR K = 1 TO 3				T		
ACCEPT PENS		9				
IF PENS > 8 AND PENS < 10 THEN					T	
CNT1 = CNT1 + 1			1			
NEXT K	4					
FOR K = 1 TO 3				F		
DISPLAY CNT1						1

Answer:

a. 1

6.3.

INSTRUCTION	K	PENS	CNT1	K <= 3	PENS >= 8 OR PENS < 10	OUTPUT
DECLARE K, PENS, TOTAL, CNT1 AS INTEGER	0	0	0			
FOR K = 1 TO 3	1			T		
ACCEPT PENS		8				
IF PENS >= 8 OR PENS < 10 THEN					T	
CNT1 = CNT1 + 1			1			
NEXT K	2					
FOR K = 1 TO 3				T		
ACCEPT PENS		10				
IF PENS >= 8 OR PENS < 10 THEN					T	
CNT1 = CNT1 + 1			2			
NEXT K	3					
FOR K = 1 TO 3				T		
ACCEPT PENS		9				
IF PENS >= 8 OR PENS < 10 THEN					T	
CNT1 = CNT1 + 1			3			
NEXT K	4					
FOR K = 1 TO 3				F		
DISPLAY CNT1						3

Answer:

c. 3

6.4.

INSTRUCTION	K	PENS	CNT1	K <= 3	PENS < 8 AND PENS > 10	OUTPUT
DECLARE K, PENS, TOTAL, CNT1 AS INTEGER	0	0	0			
FOR K = 1 TO 3	1			T		
ACCEPT PENS		8				
IF PENS < 8 AND PENS > 10 THEN					F	
NEXT K	2					
FOR K = 1 TO 3				T		
ACCEPT PENS		10				
IF PENS < 8 AND PENS > 10 THEN					F	
NEXT K	3					
FOR K = 1 TO 3				T		
ACCEPT PENS		9				
IF PENS < 8 AND PENS > 10 THEN					F	
NEXT K	4					
FOR K = 1 TO 3				F		
DISPLAY CNT1						0

Answer:

d. none of the above

QUESTION 7

7.1.

INSTRUCTION	LIGHT	VOLTS	MSG	NOT (LIGHT <> "OFF")	OUTPUT
DECLARE VOLTS AS REAL		0			
DECLARE MSG, LIGHT AS STRING	NULL		NULL		
ACCEPT LIGHT	ON				
IF NOT LIGHT <> "OFF" THEN				F	
VOLTS = 15 * 30 + 1		451			
MSG = "BRIGHT"			BRIGHT		
DISPLAY MSG, LIGHT, VOLTS					BRIGHT ON 451

Answer:

d. BRIGHT ON 451

7.2.

INSTRUCTION	LIGHT	VOLTS	MSG	NOT (LIGHT = "OFF")	OUTPUT
DECLARE VOLTS AS REAL		0			
DECLARE MSG, LIGHT AS STRING	NULL		NULL		
ACCEPT LIGHT	ON				
IF NOT LIGHT = "OFF" THEN				T	
VOLTS = 0		0			
MSG = "DARK"			DARK		
DISPLAY MSG, LIGHT, VOLTS					DARK ON 0

Answer:

d. DARK ON 0

7.3.

INSTRUCTION	LIGHT	VOLTS	MSG	NOT (LIGHT = "OFF")	OUTPUT
DECLARE VOLTS AS REAL		0			
DECLARE MSG, LIGHT AS STRING	NULL		NULL		
ACCEPT LIGHT	OFF				
IF NOT LIGHT = "OFF" THEN				F	
VOLTS = 15 * 30		450			
MSG = "BRIGHT"			BRIGHT		

DISPLAY MSG, LIGHT, VOLTS					BRIGHT OFF 450
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Answer:

c. BRIGHT OFF 450

7.4.

INSTRUCTION	LIGHT	VOLTS	MSG	NOT (LIGHT = "ON")	OUTPUT
DECLARE VOLTS AS REAL		0			
DECLARE MSG, LIGHT AS STRING	NULL		NULL		
ACCEPT LIGHT	OFF				
IF NOT LIGHT = "ON" THEN				T	
VOLTS = 100		100			
MSG = "DARK"			DARK		
DISPLAY MSG, LIGHT, VOLTS					DARK OFF 100

Answer:

a. none of these

QUESTION 8

8.1.

INSTRUCTION	MM	NUM1	NUM2	NUM3	MM ≤ 17
DECLARE MM, NUM1, NUM2, NUM3 AS REAL	0	0	0	0	
NUM1 = 5		5			
NUM2 = (NUM1 * 4 – (NUM1 – 2))			17		
FOR MM = NUM1 TO NUM2	5				T
NUM3 = NUM2 \ MM				3	
NEXT MM	6				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				2	
NEXT MM	7				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				2	
NEXT MM	8				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				2	
NEXT MM	9				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	10				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	11				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	12				

FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	13				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	14				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	15				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	16				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	17				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 \ MM				1	
NEXT MM	18				
FOR MM = NUM1 TO NUM2					F

Answer:

a. 1

8.2.

INSTRUCTION	MM	NUM1	NUM2	NUM3	MM ≤ 17
DECLARE MM, NUM1, NUM2, NUM3 AS REAL	0	0	0	0	
NUM1 = 5		5			
NUM2 = (NUM1 * 4 – (NUM1 – 2))			17		
FOR MM = NUM1 TO NUM2	5				T
NUM3 = NUM2 MOD MM				2	
NEXT MM	6				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				5	
NEXT MM	7				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				3	
NEXT MM	8				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				1	
NEXT MM	9				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				8	
NEXT MM	10				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				7	
NEXT MM	11				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				6	
NEXT MM	12				
FOR MM = NUM1 TO NUM2					T

NUM3 = NUM2 MOD MM				5	
NEXT MM	13				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				4	
NEXT MM	14				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				3	
NEXT MM	15				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				2	
NEXT MM	16				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				1	
NEXT MM	17				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 MOD MM				0	
NEXT MM	18				
FOR MM = NUM1 TO NUM2					F

Answer:

c. 0

8.3.

INSTRUCTION	MM	NUM1	NUM2	NUM3	MM ≤ 20
DECLARE MM, NUM1, NUM2, NUM3 AS REAL					
NUM1 = 6		6			
NUM2 = (NUM1 * 4 – (NUM1 – 2))			20		
FOR MM = NUM1 TO NUM2	6				T
NUM3 = NUM2 - MM				14	
NEXT MM	7				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				13	
NEXT MM	8				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				12	
NEXT MM	9				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				11	
NEXT MM	10				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				10	
NEXT MM	11				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				9	
NEXT MM	12				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				8	
NEXT MM	13				
FOR MM = NUM1 TO NUM2					T

NUM3 = NUM2 - MM				7	
NEXT MM	14				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				6	
NEXT MM	15				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				5	
NEXT MM	16				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				4	
NEXT MM	17				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				3	
NEXT MM	18				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				2	
NEXT MM	19				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				1	
NEXT MM	20				
FOR MM = NUM1 TO NUM2					T
NUM3 = NUM2 - MM				0	
NEXT MM	21				
FOR MM = NUM1 TO NUM2					F

Answer:

b. 0

8.4.

INSTRUCTION	MM	NUM1	NUM2	NUM3	MM ≤ 20
DECLARE MM, NUM1, NUM2, NUM3 AS REAL					
NUM1 = 6		6			
NUM2 = (NUM1 * 4 – (NUM1 – 2))			20		
FOR MM = NUM1 TO NUM2	6				T
NUM3 = MM - NUM2 + 1				-13	
NEXT MM	7				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-12	
NEXT MM	8				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-11	
NEXT MM	9				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-10	
NEXT MM	10				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-9	
NEXT MM	11				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-8	
NEXT MM	12				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-7	
NEXT MM	13				
FOR MM = NUM1 TO NUM2					T

NUM3 = MM - NUM2 + 1				-6	
NEXT MM	14				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-5	
NEXT MM	15				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-4	
NEXT MM	16				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-3	
NEXT MM	17				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-2	
NEXT MM	18				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				-1	
NEXT MM	19				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				0	
NEXT MM	20				
FOR MM = NUM1 TO NUM2					T
NUM3 = MM - NUM2 + 1				1	
NEXT MM	21				
FOR MM = NUM1 TO NUM2					F

Answer:

a. 1

QUESTION 9

9.1 Answer: b. False

Reason: Both comparisons are false: $1.5 > 2$ is false and $2 > 4$ is false, so the OR is false.

9.2 Answer: b. False

Reason: The statement "will display FALSE" is wrong because $1.5 < 4$ is true; the OR is true so it displays TRUE.

9.3 Answer: b. False

Reason: The condition uses HT1/HT2/HT3 while the given values are for AB1/AB2/AB3 (mismatched/undefined variables), so the tested expression is not true.

9.4 Answer: a. True

Reason: Both comparisons hold: $1.5 < 2$ is true and $2 < 4$ is true, so the AND is true.

QUESTION 10

10.1 Answer: b. False

Reason: It tests $AGE = 80$ (exactly 80) instead of > 80 , and CNT1 is not initialized before incrementing.

10.2 Answer: a. True

Reason: It correctly tests $AGE > 80$ and increments the counter ($CNT1 = 1 + CNT1$ is a valid increment).

10.3 Answer: b. False

Reason: It assigns $CNT1 = TOTAL + 1$ instead of incrementing CNT1 (wrong variable used).

10.4 Answer: b. False

Reason: It tests $AGE > 50$ and then sets $CNT1 = AGE + 1$ (stores an age-based value) instead of incrementing a counter.